

Data-Driven Trading Strategies:

Enhancing Decision-Making in the Indian Stock Market

(Master Thesis)

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30th June 2025

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1. Abstract

There is huge potential in the Indian stock markets for both retail as well as institutional investors. However, finding opportunities and executing trades of real-time value poses a challenge, especially for intraday traders who have to make split-second decisions. The tools available for technical analysis are either too narrow, generic spanning over multiple geographical regions or tailored to conditions of the Indian markets, expensive or outdated. Most platforms provide only real-time stock screening or charting but lack intelligent integrated systems that function as decision-support systems.

This thesis focuses on designing and developing an all-inclusive platform dedicated to stock analysis with multi-dimensional features embedded specially for the Indian stock markets sector. It merges historical data with real-time information to enhance the accuracy of visual analytics and predictive computations necessary to support rapid trading decisions. Core functions involve candlestick chart visualization as well as breakout signal detection using advanced machine learning techniques; volume momentum analysis per sector along with automated calculation of critical price levels inclusive of Support, resistance, key moving averages (50-day & 200 day), 52-week highs and lows.

The system retrieves high-resolution, minute-level OHLCV (Open, High, Low, Close, Volume) data using Zerodha's KiteConnect API and saves it into a PostgreSQL database. An XGBoost-based machine learning model is trained on this data to identify possible breakout patterns. The model issues buy or sell signals for each stock and timeframe based on its predictions of breakout or breakdown occurrences. These signals are displayed on candlestick charts for easier interpretation.

The system was developed with Python and Flask as the backend alongside Lightweight Charts and Plotly.js for the frontend visualization. The entire infrastructure is hosted on an AWS EC2 instance for enhanced scalability and performance while Zerodha's historical data is scraped from Visual Studio.

In addition, I have implemented sector trend analysis that calculates volume momentum of stocks within a sector. It visualizes the strength of the sector with a radial gauge chart which helps traders instantly know the gaining and losing sectors over time.

Assessment of the system using visual back-testing and accuracy prediction reveals that integrating real-time data flows, along with traditional technical indicators and machine learning forecasts, can significantly optimize both the speed and quality of automated trading decision-making processes.

Drawing a conclusion from this thesis indicates that integrated platforms provide Indian users with swift, intelligent insights for stock and trade actions where raw data is transformed into a practical usable format. It simultaneously serves as a robust foundation to further explore algorithmic trading and AI-driven financial analytics.